1 Solution — GIAM 8.4 Problem 2

1.1 Problem

Let F be the collection of all real-valued functions defined on the real line. Find an injection from $\mathbb{R}$ to F. Do you think it is possible to find an injection going the other way? In other words, do you think that F and $\mathbb{R}$ are equivalent? Explain.

1.2 Setup

The set F is

$$F = \{ f : \mathbb{R} \to \mathbb{R} \} = \mathbb{R}^{\mathbb{R}}.$$

Two facts we'll need from earlier in the chapter:

- *Cardinality of* $\mathbb{R}$*.* $|\mathbb{R}| = \mathfrak{c} = 2^{\aleph_0}$, the cardinality of the continuum.
- *Cantor's theorem.* For every set X, $|X| < |\mathcal{P}(X)|$. Equivalently, no injection $\mathcal{P}(X) \to X$ exists.

The problem has two halves: produce one specific injection, and decide whether the reverse injection exists.

1.3 Part 1 — An Injection $\mathbb{R} \hookrightarrow F$

Construction. Define $\iota : \mathbb{R} \to F$ by sending each $a \in \mathbb{R}$ to the *constant function* with value a:

$$\iota(a) = c_a, \qquad \text{where } c_a : \mathbb{R} \to \mathbb{R},\ \ c_a(x) = a \text{ for all } x \in \mathbb{R}.$$

Why it's injective. Suppose $\iota(a) = \iota(a')$, i.e. c_a and $c_{a'}$ agree at every point of $\mathbb{R}$. Evaluate at any single point — say $x = 0$:

$$a = c_a(0) = c_{a'}(0) = a'.$$

So $a = a'$, and ι is injective. ■

The real line embeds into F as the family of constant functions. (Many other embeddings work — see the remark at the end.)

1.4 Part 2 — No Injection $F \to \mathbb{R}$

The reverse injection does *not* exist. The set F is strictly larger than $\mathbb{R}$ — far larger, in fact. Here are three independent arguments, each illuminating a different aspect.

1.4.1 Argument A — F Contains a Copy of $\mathcal{P}(\mathbb{R})$

Restrict attention to the $\{0,1\}$-valued functions on $\mathbb{R}$:

$$F_0 \;=\; \{\, f : \mathbb{R} \to \{0,1\} \,\} \;\subseteq\; F.$$

Each such f is determined by — and determines — the subset $S_f = \{x \in \mathbb{R} : f(x) = 1\}$; this is just f's *characteristic function* read backwards. The correspondence $f \leftrightarrow S_f$ is a bijection

$$F_0 \;\overset{\sim}{\longleftrightarrow}\; \mathcal{P}(\mathbb{R}),$$

so $|F_0| = |\mathcal{P}(\mathbb{R})|$. By Cantor's theorem, $|\mathcal{P}(\mathbb{R})| > |\mathbb{R}|$. Since $F_0 \subseteq F$,

$$|F| \;\geq\; |F_0| \;=\; |\mathcal{P}(\mathbb{R})| \;>\; |\mathbb{R}|.$$

In the dominance ordering, $|F| > |\mathbb{R}|$, so no injection $F \to \mathbb{R}$ exists. ■

1.4.2 Argument B — Direct Diagonal Argument

Suppose, toward contradiction, that $\phi : F \to \mathbb{R}$ is an injection. Then ϕ has a partial inverse $\phi^{-1} : \phi(F) \to F$ that is a bijection. Extend it arbitrarily to a function $\psi : \mathbb{R} \to F$ (e.g., $\psi(x) = c_0$ for $x \notin \phi(F)$). Then ψ is a *surjection*: every $f \in F$ equals $\psi(\phi(f))$.

Now run Cantor's diagonal trick. Define $d : \mathbb{R} \to \mathbb{R}$ by

$$d(x) \;=\; \psi(x)(x) + 1.$$

For *any* $x \in \mathbb{R}$, the function d disagrees with $\psi(x)$ at the point x:

$$d(x) \;=\; \psi(x)(x) + 1 \;\neq\; \psi(x)(x).$$

So $d \neq \psi(x)$ for every x — i.e., d is not in the image of ψ. But $d \in F$ (it's manifestly a function $\mathbb{R} \to \mathbb{R}$), contradicting surjectivity. Therefore no injection $\phi : F \to \mathbb{R}$ exists. $\blacksquare$

1.4.3 Argument C — Cardinal Arithmetic

Just compute:

$$\begin{aligned} |F| \;&=\; |\mathbb{R}^{\mathbb{R}}| \;=\; |\mathbb{R}|^{|\mathbb{R}|} \;=\; \mathfrak{c}^{\mathfrak{c}} \\ &=\; (2^{\aleph_0})^{\mathfrak{c}} \;=\; 2^{\aleph_0 \cdot \mathfrak{c}} \;=\; 2^{\mathfrak{c}}. \end{aligned}$$

The arithmetic move $\aleph_0 \cdot \mathfrak{c} = \mathfrak{c}$ uses $\aleph_0 \leq \mathfrak{c}$ and the absorbing property of infinite cardinal multiplication ($\kappa \cdot \lambda = \max(\kappa, \lambda)$ for infinite κ, λ). Since $2^{\mathfrak{c}} > \mathfrak{c}$ by Cantor's theorem applied to $\mathbb{R}$,

$$|F| \;=\; 2^{\mathfrak{c}} \;>\; \mathfrak{c} \;=\; |\mathbb{R}|. \quad \blacksquare$$

1.5 Conclusion

F and $\mathbb{R}$ are *not* equivalent. An injection $\mathbb{R} \hookrightarrow F$ exists (the constants), but no injection $F \hookrightarrow \mathbb{R}$ does (Arguments A, B, C — pick your favorite). In the dominance ordering of Section 8.4,

$$|\mathbb{R}| \;<\; |F|,$$

and the "gap" is exactly one application of Cantor's theorem: F contains a copy of $\mathcal{P}(\mathbb{R})$, which sits one cardinality-rung above $\mathbb{R}$.

1.6 Remark — Other Injections $\mathbb{R} \hookrightarrow F$

The constant-function embedding is the cleanest, but many other maps work. Each row below is a different injection $a \mapsto f_a$:

Map $a \mapsto f_a$	**Recovery — why injective**
$f_a(x) = a$ (constant)	$f_a(0) = a$.
$f_a(x) = x + a$ (translation)	$f_a(0) = a$.

$f_a(x) = a \cdot x$ (scaling through origin)	$f_a(1) = a$.
$f_a(x) = e^{ax}$	$f_a(1) = e^a$, so $\log f_a(1) = a$.
$f_a(x) = \mathbf{1}_{\{a\}}(x)$ (indicator)	f_a has a unique nonzero point at $x = a$.

Table 1.1

The common pattern: each formula has a *parameter* a that can be read back off the function — by evaluating at a chosen point, or by inspecting where the function is nonzero. Any such recovery procedure forces injectivity.

1.7 Remark — Continuous Functions Are a Different Story

Restricting to *continuous* real-valued functions on $\mathbb{R}$ — call this set $C(\mathbb{R})$ — the cardinality drops dramatically. A continuous function is determined by its values on the rationals: if $f, g \in C(\mathbb{R})$ and $f(q) = g(q)$ for every $q \in \mathbb{Q}$, then $f = g$ on all of $\mathbb{R}$ (any real is a limit of rationals, and continuity transports the equality). So the restriction map

$$C(\mathbb{R}) \hookrightarrow \mathbb{R}^{\mathbb{Q}}, \qquad f \mapsto f|_{\mathbb{Q}}$$

is *injective*. The codomain has cardinality $\mathfrak{c}^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0} = 2^{\aleph_0} = \mathfrak{c}$. Combined with the trivial lower bound (constants alone give $|C(\mathbb{R})| \geq \mathfrak{c}$):

$$|C(\mathbb{R})| = \mathfrak{c} = |\mathbb{R}|.$$

So $C(\mathbb{R})$ *is* equivalent to $\mathbb{R}$. The "extra" $2^{\mathfrak{c}}$ functions in F that push us up a rung are precisely the *wildly discontinuous* ones — functions whose values at different points are unconstrained by any regularity.

This is a nice illustration of how *regularity hypotheses tame cardinality*: ask for measurable, Lebesgue-integrable, smooth, or even just continuous, and you stay at $\mathfrak{c}$; drop all regularity and the count immediately jumps to $2^{\mathfrak{c}}$.

1.8 Remark — The Cantor Hierarchy

Section 8.4 gives the language of *dominance*, the strict ordering on cardinal numbers. Iterating Cantor's theorem produces an infinite ladder of strict inequalities:

$$\aleph_0 \;<\; \mathfrak{c} \;<\; 2^{\mathfrak{c}} \;<\; 2^{2^{\mathfrak{c}}} \;<\; \cdots$$

Concrete representatives of each rung:

Rung	**Cardinal**	**Example sets**
0	$\aleph_0$	$\mathbb{N}, \mathbb{Z}, \mathbb{Q}$
1	$\mathfrak{c}$	$\mathbb{R}, \mathbb{R}^n, C(\mathbb{R})$
2	$2^{\mathfrak{c}}$	$\mathcal{P}(\mathbb{R}), \quad F = \mathbb{R}^{\mathbb{R}}$
3	$2^{2^{\mathfrak{c}}}$	$\mathcal{P}(F)$

Table 1.2

This problem places F on rung 2 — strictly above $\mathbb{R}$, strictly below $\mathcal{P}(F)$. Dominance lets us *talk* about this hierarchy precisely, and Cantor's theorem guarantees the ladder never collapses.

(The *Continuum Hypothesis*, treated in Section 8.5, asks whether any cardinality sits *between* $\aleph_0$ and $\mathfrak{c}$; Gödel and Cohen showed this question is independent of ZFC. The analogous question for cardinalities between $\mathfrak{c}$ and $2^{\mathfrak{c}}$ — the *generalized* continuum hypothesis — is similarly independent.)